

Solving Optimal Electric Vehicle Charging Station Placement Problem Using Digital Quantum Annealing

Chia-Ho Ou, Chung-Chieh Cheng, Chih-Yu Chen, Krischonme Bhumkittipich, and Sillawat Romphochai

Abstract—It is imperative to reduce carbon dioxide emissions and fossil fuel consumption to ensure the Earth's sustainable future and mitigate negative impacts like extreme climate changes caused by greenhouse gases. Electric vehicles (EVs) play a pivotal role in this context, where the convenience of charging stations is crucial in influencing consumer choices. This paper formulates the placement of charging stations as an optimization problem, considering factors like station density, accessibility, coverage, and regional cost variations, to minimize total construction costs. The problem is mapped onto a quadratic unconstrained binary optimization (QUBO) model. The QUBO can efficiently represent and solve complex combinatorial optimization problems, making it suitable for quantum and advanced classical algorithms. Utilizing digital quantum annealing, the optimal or near-optimal solutions are efficiently identified. The performance of the proposed approach is compared with classical algorithms, including mixed linear programming and simulated annealing, demonstrating significant advancements of digital annealing.

Index Terms—Digital annealing, optimization, quantum annealing, quantum application.

I. INTRODUCTION

THE United Nations and many countries have issued declarations and policies to reduce the use of fuel-powered vehicles. At the 26th United Nations Climate Change Conference (COP26), over 30 countries and six significant automakers signed an agreement to complete road vehicle electrification by 2040 [1]. In Taiwan, greenhouse gas emissions from the transportation sector account for about 37 $MtCO_2e$, representing 12.8% of the country's total emissions. The majority of these emissions come from road transportation. Consequently, in 2022, the Taiwanese government announced a ban on the sale of fuel-powered motorcycles and cars by 2040 [2], leading to a rapid increase in electric vehicles (EVs) [3]. The construction of charging stations is thus essential, especially given Taiwan's high population density, making the location of these stations particularly essential. As EVs have not yet entirely replaced fuel vehicles and their

charging times are longer, charging stations can become saturated, leading to longer waiting times. Therefore, traditional gas stations cannot be completely replaced. Many countries are still in a transitional period regarding the construction of charging stations, hence the progress in building these stations has been relatively slow. Additionally, the limited range of EVs significantly reduces their practicality and increases range anxiety among owners. In many cases, partnerships with parking lots or shopping malls are sought to build charging piles, lower costs and attract foot traffic to these locations, thereby enhancing the charging experience for vehicle owners. The selection of locations for charging stations and the formation of a charging station network are thus of great importance.

In the quantum computing field, quantum computers can be divided into general-purpose and task-specific quantum computers. General-purpose quantum computers are based on quantum bits (i.e., qubits) and can perform various computational tasks, similar to classical computers. Task-specific quantum computers, such as quantum annealers, are designed for specific computational tasks. Digital quantum annealing, is inspired by thermal annealing in metallurgy. Kirkpatrick et al. proposed a probabilistic algorithm called simulated annealing (SA) [4], which seeks global optimal or near-optimal solutions in combinatorial optimization problems. Digital annealing is primarily based on the simulated annealing and is inspired by the techniques of quantum annealing. It is suitable for solving complex combinatorial problems [5]–[9], such as NP-hard problems, which require exponentially growing computation times in classical domains.

Digital annealing can significantly reduce these computation times and find more optimal solutions, increasing its practicality. With the increasing maturity of modern digital annealing techniques and their applications in commercial cases in finance and logistics [10], [11], Fujitsu's third-generation digital annealer (DA3) [12] is used to map the problem into a quadratic unconstrained binary optimization (QUBO) model. This approach transforms constrained cost functions into binary quadratic polynomial representations in spin Hamiltonians for problem-solving. This enables rapid calculation of optimal or near-optimal solutions in complex and changing environments, overcoming the limitations of classical computers in solving such problems. This paper considers various constraints and formulates the location problem of charging stations as an optimization model, to find an optimal solution under these constraints. This paper uses digital annealing to solve the problem and compare its performance with classical

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methods like mixed linear programming and simulated annealing.

The reason for employing the QUBO model rather than traditional methods lies primarily in its compatibility with quantum and digital quantum annealing algorithms. Traditional optimization methods, such as mixed integer linear programming (MILP) and heuristic algorithms, typically encounter scalability and efficiency issues as the size of the optimization problem grows significantly. On the other hand, the QUBO formulation naturally aligns with the architecture of quantum-inspired and quantum computing platforms, particularly quantum annealers and digital annealers, providing substantial computational advantages for large-scale, complex combinatorial problems. Specifically, digital quantum annealing, such as Fujitsu's digital annealer, leverages quantum-inspired parallel exploration and tunneling effects, allowing rapid exploration of complex energy landscapes. This approach has demonstrated significantly better performance in terms of runtime and scalability compared to classical optimization techniques. Thus, applying the QUBO formulation in EV charging station placement problems, which involve numerous binary decision variables and complicated interdependencies among constraints, is particularly advantageous. It not only ensures computational tractability but also achieves better optimality and faster convergence in realistic scenarios involving large numbers of candidate locations and constraints.

It should be explicitly noted that the digital annealing method employed in this study is a quantum-inspired computational paradigm rather than a quantum technology in the strict sense. Digital annealing leverages concepts analogous to quantum tunneling and parallel exploration strategies inspired by quantum annealing. Although digital annealing does not operate using quantum states or quantum hardware, it provides significant practical advantages in solving large-scale combinatorial optimization problems efficiently on classical hardware. Nevertheless, the formulation of our optimization model as a QUBO problem inherently ensures compatibility with genuine quantum annealing hardware, such as D-Wave quantum annealers [13], thereby enabling seamless future integration with real quantum technologies. Hence, this study serves not only as a practical and immediate solution to real-world challenges but also lays a foundational model for subsequent implementation on actual quantum hardware, reflecting its broader significance within the quantum computing landscape.

The structure of the paper is as follows: Section II covers related work, Section III presents the proposed problem model, Section IV outlines the proposed solutions, Section V discusses the experiments results and discussion, and Section VI concludes the paper.

II. RELATED WORK

With the advancement of EVs, charging stations have become a widely studied topic. Regarding charging station location, most studies propose optimization models from various perspectives. The complexity of these problems has been proved to be NP-hard [14]. Several methods addressing this

issue have been proposed [15], adopting a traffic flow model similar to that in [16], where a source node O^i connects to node i , with n units of flow entering from the source node. This approach ensures the connectivity of the charging station network, allowing EVs to travel to every region. Ref. [17] proposes two greedy algorithm-based acceleration methods, lazy greedy with direct gain (LGDG) and lazy greedy with effective gain (LGEG), to solve charging station placement and charger allocation problems and compare their performance with general greedy algorithms and mixed linear programming.

Some studies have considered the convenience of EV owners. Ref. [18] considers road traffic conditions and charging queue times, proposing the optimizing electric vehicle charging station (OCEAN) algorithm. Ref. [19] uses extensive data collection and regression analysis to predict EV owners' potential parking times and locations for minimizing their travel costs. Economic benefits have also been a focus, Ref. [20] is presented to reduce maintenance costs, network losses, and improve voltage curves to minimize overall costs. Ref. [21] considers annual comprehensive operating costs and proposes the adaptive particle swarm optimization (APSO) algorithm. Ref. [22] takes into account the costs of voltage regulation and protection devices to increase economic benefits. Ref. [23] examines economic factors, electricity prices, and EV types for EV charging infrastructure with simulations based on local conditions in Indonesia.

Although charging stations have not yet fully replaced gas stations, Jia et al. [24] optimized the location and size of charging stations based on user behavior and charging distribution by utilizing principles of reusing existing gas stations. Many studies focus on optimizing from a technical perspective, such as [25], which reduces network losses and improve voltage curves while studying the relationship between load and generation. Ref. [26] focuses on node voltage deviation and power loss to maximize charges, using the cross-entropy method for planning models.

Ajagekar and You [27] explored the use of quantum computing for energy system planning, comparing it with traditional computing to achieve better solutions in less time. They also noted that not all optimization problems are suitable for quantum computing. Ref. [28] allocates charging piles and power based on EV information, using quantum computing for scheduling and power distribution control for EVs on highways, reducing charging queue times and extending battery state and lifespan. Ref. [29] used quantum annealing techniques to study the charging of plug-in hybrid electric vehicles, optimizing power distribution at charging stations to reduce queue times and energy loss. This paper does not conduct in-depth research on smart grids [30], [31] or other technical aspects and queue scenarios, focusing on applying these methods flexibly in various scenarios. The factors such as the convenience, coverage, and charging demand of the charging network, as well as regional differences are considered to enable EV drivers to travel freely to any location. Our paper focuses on optimizing the location of charging stations, believing that sufficient charging stations and rapid charging speeds are essential to solving range anxiety and convenience

issues while reducing costs and meeting user needs.

Existing research on EV charging station placement primarily relies on classical algorithms to address a range of issues, such as optimizing the stations' locations, density, and accessibility, in line with traffic flow, user convenience, and economic considerations. These studies provide essential insights into algorithm efficiency and the practical needs for creating effective EV charging networks, with methods like greedy algorithms and mixed linear programming aiming to balance construction costs against the necessity for broad accessibility to encourage EV use. Meanwhile, quantum computing research in the EV sector has focused more on optimizing charging behavior, such as managing and scheduling to enhance efficiency and battery life or reduce wait times, taking advantage of quantum algorithms' ability to handle complex problems more adeptly than classical methods. Yet, this body of work has not applied quantum solutions to the specific issue of charging station placement.

Our paper fills this gap by applying digital annealing techniques to the EV charging station placement problem, marking a pioneering approach in the literature. It integrates the logistical and spatial optimization strengths of classical algorithms with the potential of quantum computing to improve EV charging behavior, applying a digital annealing algorithm for the strategic deployment of charging stations. This novel integration not only leverages the superior processing power of quantum algorithms but also addresses both the behavioral and logistical aspects of EV charging, offering a comprehensive and forward-looking solution to the infrastructure challenges of supporting an expanding EV market.

Recent research efforts in EV charging station placement have explored using quantum computing. For instance, Sakib and Chen [32] proposed a quantum annealing-based mathematical model to optimize EV charging infrastructure placement with the primary goal of maximizing service quality for EV users. Their approach was validated using real-world scenarios solved by commercial quantum annealers, demonstrating rapid convergence and solution insensitivity to variations in deployment areas.

Radvand et al. [33] introduced a quantum optimization algorithm specifically tailored for intercity EV charging station placement. Utilizing quantum parallelism and amplitude amplification, their algorithm showed significant computational improvements over classical exact methods, efficiently addressing the inherent complexity of intercity network optimization problems.

Additionally, Chandra et al. [34] explored a hybrid optimization algorithm combining quantum annealing and genetic algorithms, showcasing significant improvement in optimization performance. Their hybrid approach substantially reduced the minimal distance from points of interest compared to quantum annealing alone, illustrating the potential of hybrid methodologies in enhancing algorithmic efficiency.

While these recent works have demonstrated significant advances in leveraging quantum computing techniques and insights for EV charging placement optimization, our study distinguishes itself in several critical aspects. Unlike previous studies that primarily focus on algorithmic efficiency or single-

dimensional objectives, our research emphasizes integrating multiple practical constraints such as coverage efficiency, accessibility, cost effectiveness, and urban environmental impact into a comprehensive optimization framework. Moreover, our method utilizes digital quantum annealing, a quantum-inspired approach specifically suited for practical and scalable deployment on currently available hardware, making it highly relevant and immediately applicable for real-world urban planning scenarios.

III. PROPOSED PROBLEM FORMULATION

The EV charging station placement problem is inherently combinatorial and complex due to numerous candidate locations and interdependent constraints (coverage, cost, and demand). Classical methods, such as MILP, simulated annealing (SA), and other heuristic algorithms, typically suffer from computational inefficiency and limited scalability. The QUBO formulation is highly suitable for this specific EV charging station placement issue as it naturally maps the decision-making process into binary variables and quadratic interactions, perfectly aligning with quantum-inspired computation platforms such as digital quantum annealing. By leveraging parallel exploration capabilities, digital annealing rapidly navigates through the vast combinatorial search space, significantly improving solution quality, convergence speed, and scalability. Such characteristics make QUBO not only advantageous but also uniquely suitable for addressing the EV charging station placement problem.

A. System Model

In this paper, the problem of charging pile deployment as a combinatorial optimization problem is described to consider various constraints such as construction costs, charging demand, coverage, etc. The problem is then converted into a QUBO form. The QUBO model is considered foundational in the field of quantum annealing and digital annealing. It is found that QUBO is effective in solving many significant combinatorial optimization problems [35]–[37]. The QUBO model has also been shown to play a prominent role in physics, particularly in the Ising model [38], which is currently one of the most widely used optimization models in the quantum domain. The problems are solved by mapping them into a QUBO model and processing them in a quantum computer (i.e., annealer).

The digital annealing requires the problem to be mapped into an energy expression of the Hamiltonian operator, typically denoted as H . The objective is to find the variable combination that minimizes the value of H , which may be a polynomial of degree n . For ease of mathematical computation, the QUBO model is used for this transformation. The formal definition of the QUBO model is shown as

$$QUBO : \text{minimize } y = x^T Q x \quad (1)$$

Here, x is a vector of binary decision variables, Q is a constant square matrix also known as the QUBO matrix, and y is the objective to be minimized.

TABLE I
MODEL PARAMETERS.

G	An undirected graph
V	Set of candidate locations for charging station construction
E	Set of edges between charging stations
$\bar{G}$	An undirected graph
$\bar{V}$	Set of points with edge lengths less than or equal to R and greater than or equal to αR
$\bar{E}$	Set of edges with lengths less than or equal to R and greater than or equal to αR
n	Number of candidate locations for charging station construction
x_i	Set of decision variables for charging stations
C_i	Cost at point i
R	Average driving range of electric vehicles
α	Discount coefficient
$d(i, j)$	Distance from charging station i to charging station j
s_i	Service range of the charging station at point i
D_i	Charging demand at point i
f_i	Charging power at point i
$CityArea$	City area
T	Terminal nodes
$V_j^{\alpha R}$	Set of charging stations within a distance less than R and greater than αR from charging station j

In practical scenarios, most problems must satisfy additional constraints. By introducing quadratic penalties in the objective function as an alternative to classical constraints, constrained models can be effectively remapped into the form of QUBO [39]. This approach enables efficient problem-solving even when there are hundreds to thousands of variables and constraints [40]. The penalty parameter is a positive energy penalty value, which must be chosen sufficiently large to ensure that the penalty term effectively equals classical constraints. Different values can be given to each different constraint, with larger penalty parameter values indicating stricter constraints. In our model, the parameters is listed in Table I.

In this paper, an undirected graph $G = (V, E)$ is utilized to simulate the network of charging stations. Here, V represents the candidate locations for constructing charging stations, and E represents the connections between these candidate locations. To ensure that EVs can reach the next charging station, the shortest path distance $d(i, j)$ from node i to j must be less than or equal to the average driving range R of most EVs on the market. Therefore, E is defined as $E = \{(i, j) | i, j \in V, d(i, j) \leq R, i \neq j\}$. The average charging capacity f_i at node i is related to the size of the site and the surrounding traffic conditions. Each node also has a local average charging demand D_i , which is correlated with the number of EVs in that area; the more EVs there are, the larger the value of f_i .

The value of C_i represents the construction cost of node i , including land price, rent, site construction, etc. These costs can be estimated based on the flow of people and population density. s_i is the service radius of the charging station, related to the nearby traffic flow and the convenience of transportation. T represents the set of terminal nodes. Our goal is to construct charging stations in remote locations with fewer EVs.

B. Problem Formulation

To address the issue of charging stations being overly concentrated, a discount coefficient $\alpha \in \{0, 1\}$ is defined. The undirected graph G is redefined as $\bar{G} = (\bar{V}, \bar{E})$, where:

$$\bar{V} = \{j \in V | \alpha R \leq d(i, j) \leq R, i \neq j\} \quad (2)$$

$$\bar{E} = \{(i, j) | i, j \in V, \alpha R \leq d(i, j) \leq R, i \neq j\} \quad (3)$$

The proposed EV charging station placement problem is formulated mathematically as follows:

The decision variable x_j is defined as a binary variable representing whether a charging station is constructed at location j :

$$x_j = \begin{cases} 1 & \text{if a charging station is constructed at} \\ & \text{candidate location } j, \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

The primary objective is to minimize the total construction cost, given by:

$$\text{minimize} \quad \sum_{j=1}^n C_j x_j, \quad \forall j \in V \quad (5)$$

where C_j denotes the construction cost at candidate location j , including factors such as land acquisition, construction expenses, and operational costs.

This optimization problem is subject to the following constraints:

- **Coverage constraint:** ensures that EVs can reach a subsequent charging station within their driving range:

$$x_i + x_j \geq 1, \quad \forall (i, j) \in E, \quad (6)$$

which guarantees at least one charging station is constructed at either end of each edge (i, j) within the feasible driving range.

- **Demand satisfaction constraint:** ensures each charging station can satisfy the total charging demand (D_i) of the surrounding EV population:

$$\sum_{i \in V_j^{\alpha R}} f_i x_i \geq D_j, \quad \forall j \in V, \quad (7)$$

where f_i represents the average charging capacity at node i , and $V_j^{\alpha R}$ is the set of stations within a distance between αR and R from station j . This ensures sufficient power capacity to satisfy local EV charging demand at every selected location.

- **City-wide coverage constraint:** ensures the constructed charging stations collectively cover the entire target urban area:

$$\sum_{j=1}^n s_j x_j \geq \text{CityArea}, \quad (8)$$

where s_j denotes the service coverage area of a charging station at location j , considering traffic flow and accessibility. This guarantees comprehensive urban area coverage by selected charging stations.

- **Remote area constraint:** ensures at least one charging station is constructed in remote areas, where EV density is typically lower:

$$\sum_{j \in T} x_j \geq 1, \quad (9)$$

where T is the set of identified remote or peripheral candidate locations requiring specific coverage considerations.

The complete problem formulation is shown as follows.

Objective: Minimize the total cost of constructing charging stations:

$$\text{minimize } \sum_{j=1}^n C_j x_j, \forall j \in \bar{V}. \quad (10)$$

Subject to:

$$x_i + x_j \geq 1, \forall i, j \in \bar{E}, \quad (11)$$

$$\sum_{i \in V_j^{\alpha R}} f_i x_i \geq D_i, \forall i, j \in \bar{E}, \quad (12)$$

$$\sum_{j=1}^n s_j x_j \geq \text{CityArea}, \forall j \in \bar{V}, \quad (13)$$

$$\sum_{j \in T} x_j \geq 1. \quad (14)$$

Subsequently, the constrained optimization problem is reformulated into the QUBO model by incorporating these constraints as penalty terms in the objective function. This transformation enables effective and efficient computation using quantum-inspired optimization algorithms such as digital annealing.

IV. PROPOSED DIGITAL ANNEALING SCHEME AND COMPARED CLASSICAL SOLUTIONS

In this paper, three different solution methods are employed: digital annealing, simulated annealing, and mixed linear programming. Both digital annealing and simulated annealing require the transformation of the model into a QUBO model for problem-solving.

A. Proposed Digital Annealing Scheme

The digital annealer, DA3, offers a specialized method for handling linear inequalities, eliminating the need for additional slack variables. This significantly reduces the extra variables introduced by slack variables. The penalty and cost terms are separated and uploaded to Fujitsu's Digital Annealing Cloud Service for problem solving. The DA3 conducts searches in the solution space with features like automatic temperature control, generation of starting points for searches, and automatic adjustment of penalty coefficients. The digital annealing process to simulate quantum tunneling to accelerate the search for global or local minimum energy. The problem formulation for DA3 is thus derived as

$$\text{minimize } \sum_{j=1}^n C_j x_j, \forall j \in \bar{V}, \quad (15)$$

Subject to:

$$-x_i - x_j \leq -1, \forall i, j \in \bar{E}, \quad (16)$$

$$-\sum_{i \in V_j^{\alpha R}} f_i x_i \leq -D_i, \forall i, j \in \bar{E}, \quad (17)$$

$$-\sum_{j=1}^n s_j x_j \leq -\text{CityArea}, \forall j \in \bar{V}, \quad (18)$$

$$-\sum_{j \in T} x_j \leq -1. \quad (19)$$

B. Simulated Annealing (SA) Scheme

The SA scheme randomly finds a low point as a starting point, and at a local minimum energy, there is a probability to overcome the peak to find the global minimum energy. This probability is proportional to $\exp(-\beta H)$, where H represents the Hamiltonian, and β is inversely proportional to temperature. The temperature gradually decreases to find the minimum energy, hence reducing the probability of overcoming peaks and converging to the minimum energy.

In the QUBO model, inequalities are assumed to be less than or equal to, so transforming inequalities in the model to equalities is the first step. Additionally, our goal is to make the penalty term equal to zero. Therefore, slack variables have to be added to the inequalities to convert them into equalities. The problem formulation for SA is shown as

$$\text{minimize } \sum_{j=1}^n C_j x_j, \forall j \in \bar{V}, \quad (20)$$

Subject to:

$$-x_i - x_j + u_l + 1 = 0, \forall i, j \in \bar{E}, u_l \in [0, 1], \quad (21)$$

$$-\sum_{i \in V_j^{\alpha R}} f_i x_i + v_l + D_i = 0, \forall i, j \in \bar{E}, v_l \in [0, \sum_{i \in V_j^{\alpha R}} f_i], \quad (22)$$

$$-\sum_{j=1}^n s_j x_j + w_l + \text{CityArea} = 0, \forall j \in \bar{V}, w_l \in [0, \sum_{j=1}^n s_j], \quad (23)$$

$$-\sum_{j \in T} x_j + z_l + 1 = 0, z_l \in [0, 1]. \quad (24)$$

C. Mixed-Integer Linear Programming (MILP) Scheme

The MILP computational method uses the GNU linear programming kit (GLPK) and Pyomo [41], both open-source optimization tools widely employed for solving linear and mixed-integer programming problems efficiently.

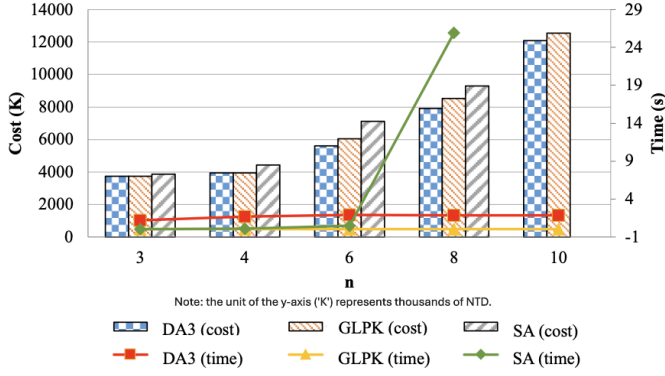


Fig. 1. Small-scale scenario: construction cost and run time.

V. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, the simulations on a small scale with 3 to 10 potential charging station locations and on a large scale with 50 to 300 locations are conducted. The performance of three methods is evaluated and compared based on the values of objective function, the run time, and the number of constructed charging stations, ensuring that constraints are not violated. The locations and parameters for potential charging stations were randomly generated using a uniform distribution within the specified geographical boundaries. Each potential location's associated parameters (cost, coverage radius, charging demand) were also uniformly sampled within realistic predefined ranges derived from urban planning data. The simulations are then conducted for both small-scale and large-scale scenarios to evaluate the performance of the three compared methods. Additionally, Taipei City and New Taipei City in Taiwan are chosen for the experimental field for further assessing the performance of the digital annealing mechanism.

Constructing the Hamiltonian and QUBO matrix can be challenging, particularly for large-scale problems. Thus, we employ PyQUBO [42], a Python library designed for intuitive and efficient QUBO model formulation, widely adopted in quantum annealing applications.

A. Simulation Results

For the small-scale scenarios, we set the driving range R to 250 km, α to 0.05, and $CityArea$ to approximately 100 km². In the small-scale scenario, for example, with an average driving limit of 400 km, the driving distance R for EVs is set to 250 km. Setting the α value to 0.05 ensures better construction of charging stations. The $CityArea$ value is set to the size of 1 to 2 administrative districts, approximately 100 km².

In the small-scale simulation results, twenty different initial placements for each number of charging station locations are conducted and the mean value is calculated as the result. Since SA has a probabilistic nature, for each test, SA is repeated running 1000 times and selected the first solution that satisfied all constraints, as shown in Table II. In Fig. 1, it is observed that as the number of locations n increases, the number of slack variables also increases, leading to a significant growth of the run time for SA. Moreover, when

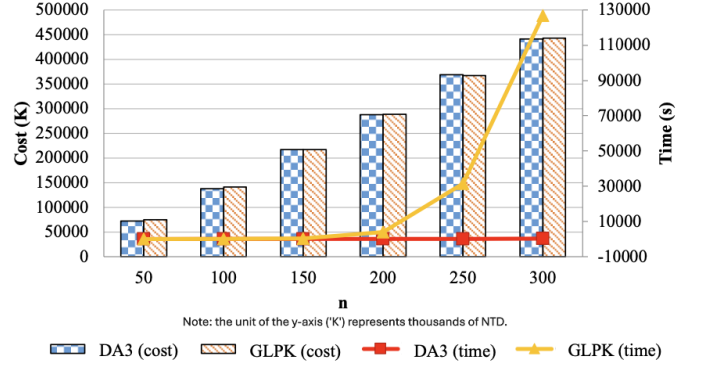


Fig. 2. Large-scale scenario: construction cost and run time.

n increases to 10, SA fails to satisfy all constraints and find a solution, and it also incurs the highest cost, indicating that SA produces the worst solutions. On the other hand, GLPK demonstrates the best performance in terms of run time, which is not only significantly short but also shows a steady growth. In small-scale scenarios, DA3 also shows a steady growth in run time. However, DA3 achieves the lowest construction cost, indicating that it provides the most optimal solutions among the three methods.

In the large-scale scenario, the value of $CityArea$ is expanded to 2323 km² with the same driving range and α value. In the large-scale simulation tests, different quantities of locations n are conducted for performance evaluations. As the number of locations is increased, the time required to find a solution is also increased significantly. Since SA was unable to find a solution satisfying all constraints when n was 10, only DA3 and GLPK are compared in the large-scale scenarios, as shown in Table III.

From Fig. 2, it is evident that DA3 generally offers slightly better solutions than GLPK. Notably, when the number of locations n exceeds 150, the time required for GLPK to find a solution shows an explosive growth. In contrast, despite a significant increase in the number of locations, DA3 still solves the problem very quickly, and its run time growth curve is quite steady. It indicates that DA3 is more suitable than GLPK for solving large-scale problems.

In Figs. 3 and 4, the simulations are conducted under different α values to observe their effects on the model. It is evident that the smaller the α value, the more edges are likely to be chosen. According to (7), more selected edges indicates a greater number of charging stations needed to be constructed. Thus, it will cause a burden on the power system. Therefore, by controlling the value of α , the density of charging stations can be regulated.

Figs. 3 and 4 also show that when the number of charging stations remains constant and the value of α is decreased, as the number of construction sites chosen by GLPK increases, the time required for solving the problem also increases. However, DA3's run time is not affected. This indicates that DA3 maintains a consistent performance in terms of run time, regardless of the number of locations selected, particularly for larger-scale problems where the number of potential sites for charging stations is high.

TABLE II
SMALL-SCALE SCENARIO: SIMULATION RESULTS.

n	# of constructed stations			Cost			Time (sec)		
	DA3	GLPK	SA	DA3	GLPK	SA	DA3	GLPK	SA
3	2.45	2.45	2.45	3,729,897.2	3,729,897.2	3,874,592.15	1.199	0.035	0.028
4	3.05	3.05	3.15	3,950,441.9	3,950,441.9	4,438,665.55	1.674	0.035	0.121
6	4.25	4.8	4.95	5,616,699.7	6,047,541.3	7,114,677.75	1.911	0.034	0.46
8	6.1	6.7	6.6	7,908,507.2	8,518,121.6	9,290,679.3	1.87	0.037	25.92
10	7.9	8.2	-	12,087,974.0	12,538,290.7	-	1.87	0.038	-

TABLE III
LARGE-SCALE SCENARIO: SIMULATION RESULTS.

n	# of constructed station		Cost		Time (sec)	
	DA3	GLPK	DA3	GLPK	DA3	GLPK
50	46.4	47.5	72,539,012	75,160,424	3.95	0.098
100	92.6	93.5	137,959,101	140,843,654	8.655	19.377
150	141	141.25	216,777,186	217,031,164	24.161	333.423
200	189	189.6	288,107,996	288,925,318	53.594	4055.54
250	237.6	237.4	368,539,113	367,432,557	120.71	31558.2
300	289.2	288.6	441,293,950	443,010,367	256.08	126706

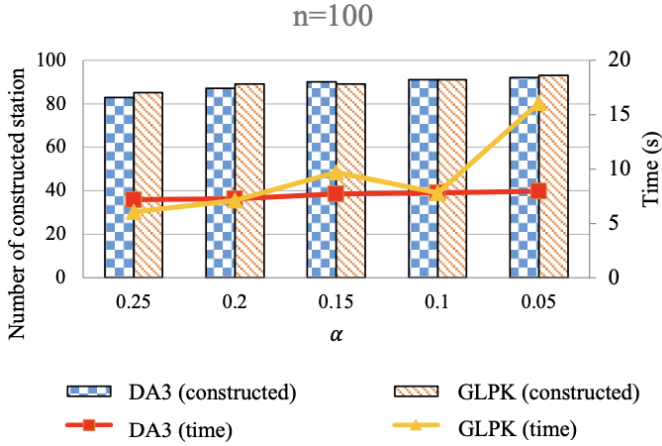


Fig. 3. Number of constructed station and run time for 100 candidate locations under varying α values.

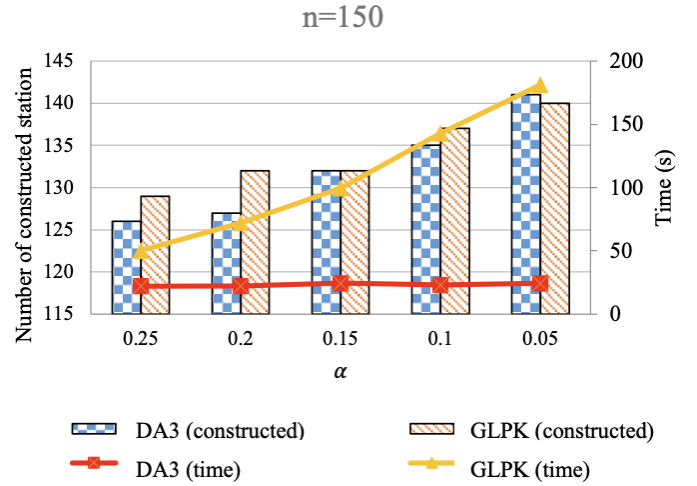


Fig. 4. Number of constructed station and run time for 150 candidate locations under varying α values.

B. Real-World Experiments

In this section, the proposed model is applied to a real-world environment. It can be seen how the model proposed in this paper can be used and implemented with the digital annealer (DA3) to achieve a sustainable country. New Taipei City, which has the highest number of EVs in Taiwan [43], and Taipei City, surrounded by New Taipei City, which has the third-highest number of EVs are chosen as the experimental field. Taipei City and New Taipei City consist of 41 administrative districts. In each district, about 2 to 3 candidate locations were chosen as potential sites for constructing charging stations [44]. Thus, there are 100 potential charging station locations in total (see Fig. 5).

As shown in Fig. 5, the number of EVs in each administrative district [45] is used to allocate the demand D_i ; the higher the number of EVs, the higher the D_i . The construction cost C_j was determined based on population density [46]; higher

population density resulted in higher C_j . Traffic flow near the charging stations [47], [48] was used to set s_j ; greater traffic flow near the charging station led to larger s_j values. The detailed parameters for potential charging stations are displayed in Table IV.

This approach allows us to realistically model the need and placement of EV charging stations in Taipei City and New Taipei City, taking into account various factors such as the density of electric vehicles, population, and traffic flow to optimize the location and number of charging stations needed. The experiments with different α values for the Taipei City and New Taipei City areas are conducted, as detailed in Table . The *CityArea* refers to the combined area of New Taipei City and Taipei City. The experimental results are presented in Figs. 6– 10.

The orange marker represents the selection for building

TABLE IV
CHARGING STATION PARAMETERS FOR TAIPEI CITY AND NEW TAIPEI CITY.

Site	D_i	C_j			s_j		
1 & 42 & 83	3,037	2,066,517	1,865,506	1,711,406	56	35	47
2 & 43 & 84	1,482	1,833,409	1,833,409	1,604,803	65	37	51
3 & 44 & 85	2,106	240,585	2,216,294	1,826,316	59	25	47
4 & 45 & 86	1,664	1,567,501	1,865,506	1,886,286	62	29	43
5 & 46 & 87	1,701	1,964,152	1,322,141	1,455,045	64	40	48
6 & 47	834	2,103,265	2,149,012	-	61	26	-
7 & 48	1,115	1,957,999	1,954,111	-	40	20	-
8 & 49 & 88	1,501	822,378	868,923	685,853	46	21	34
9 & 50 & 89	1,815	521,426	597,048	656,485	41	35	26
10 & 51 & 90	2,127	870,545	888,395	854,727	68	38	36
11 & 52 & 91	1,727	426,864	487,350	551,296	44	26	25
12 & 53 & 92	1,410	423,785	413,533	488,340	40	20	50
13 & 54 & 93	4,888	2,387,124	2,169,843	2,047,896	75	52	22
14 & 55 & 94	2,818	2,336,752	2,139,863	1,932,631	73	53	33
15 & 56 & 95	3,242	2,012,412	2,210,263	1,639,084	75	59	31
16 & 57 & 96	1,743	3,746,242	2,718,368	1,928,117	76	20	43
17 & 58 & 97	3,059	2,138,085	1,996,598	1,539,094	39	26	23
18 & 59 & 98	1,838	250,728	584,285	373,406	39	38	24
19 & 60	1,197	544,595	615,208	-	27	34	-
20 & 61	391	419,356	387,524	-	28	21	-
21 & 62	615	160,353	169,879	-	77	69	-
22 & 63	1,173	126,954	161,891	-	79	63	-
23 & 64	1,344	129,169	110,142	-	71	61	-
24 & 65	109	100,530	110,632	-	71	65	-
25 & 66 & 99	1,844	180,874	131,947	172,115	76	72	26
26 & 67 & 100	1,438	2,693,567	1,849,822	1,529,419	50	48	44
27 & 68	614	126,357	171,476	-	53	45	-
28 & 69	594	140,227	133,544	-	50	49	-
29 & 70	724	124,006	167,988	-	52	41	-
30 & 71	106	111,490	113,247	-	54	48	-
31 & 72	34	100,050	115,817	-	51	45	-
32 & 73	22	100,038	165,291	-	33	28	-
33 & 74	110	153,360	116,921	-	37	23	-
34 & 75	37	132,120	151,895	-	34	27	-
35 & 76	200	199,043	252,218	-	67	60	-
36 & 77	12	100,059	136,295	-	36	32	-
37 & 78	78	100,054	102,048	-	33	31	-
38 & 79	48	111,058	116,618	-	78	67	-
39 & 80	66	148,917	192,077	-	76	70	-
40 & 81	63	133,123	124,871	-	51	45	-
41 & 82	27	100,019	100,105	-	52	56	-

TABLE V
REAL-WORLD EXPERIMENTS RESULTS UNDER VARYING α VALUES.

α	DA3		
	# of constructed station	Cost	Time (sec)
0.25	86	66,416,834	12.217
0.2	87	70,547,961	12.116
0.15	91	75,890,075	13.505
0.1	94	81,604,928	13.794
0.05	95	83,894,144	14.463

a charging station, while other colors indicate no selection. These experiments allow us to evaluate how varying α values impact the distribution and number of charging stations needed in the combined area of Taipei City and New Taipei City. By analyzing these results, insights into the optimal deployment of charging stations in urban environments can be gained. This approach is crucial for cities planning for transition to more sustainable transportation options and supports the

infrastructure development needed for EVs.

In addition to evaluating construction cost and computational efficiency, we further analyzed the coverage efficiency based on actual service areas (s_j) from Table V across various α values. The results clearly demonstrate high coverage efficiencies across all tested α values:

- For $\alpha = 0.25$, the total coverage is approximately 3893 km^2 (168% of the target urban area).
- For $\alpha = 0.20$, coverage reaches 3927 km^2 (169%).
- For $\alpha = 0.15$, coverage is 4117 km^2 (177%).
- For $\alpha = 0.10$, coverage is 4213 km^2 (181%).
- For $\alpha = 0.05$, the highest coverage efficiency is achieved, covering 4291 km^2 (185%).

These results indicate that our proposed model consistently provides extensive coverage well beyond the actual urban area size (2323 km^2), ensuring comprehensive EV charging service availability.

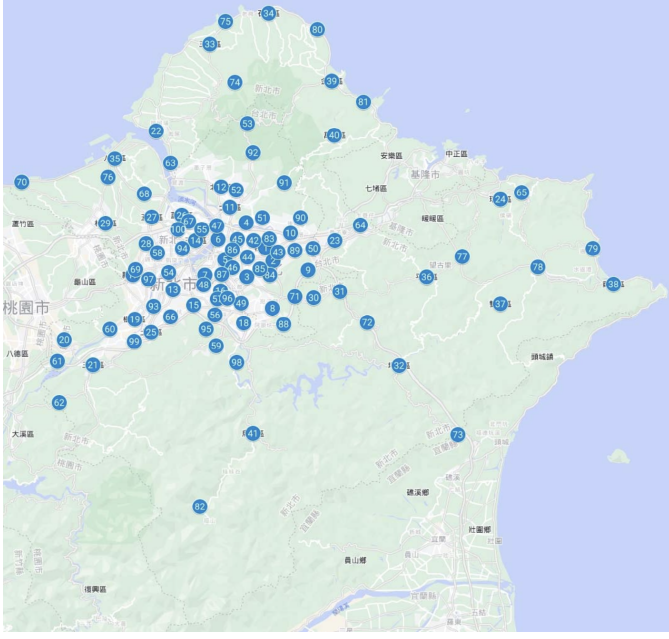
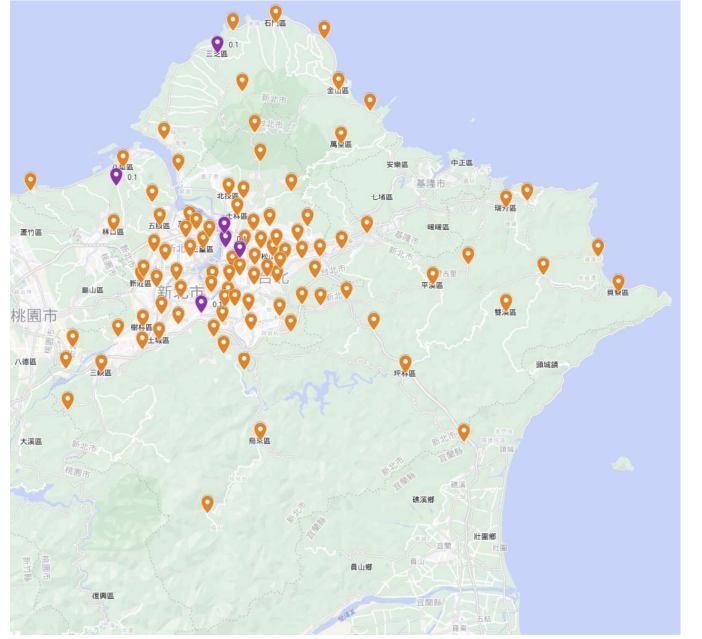
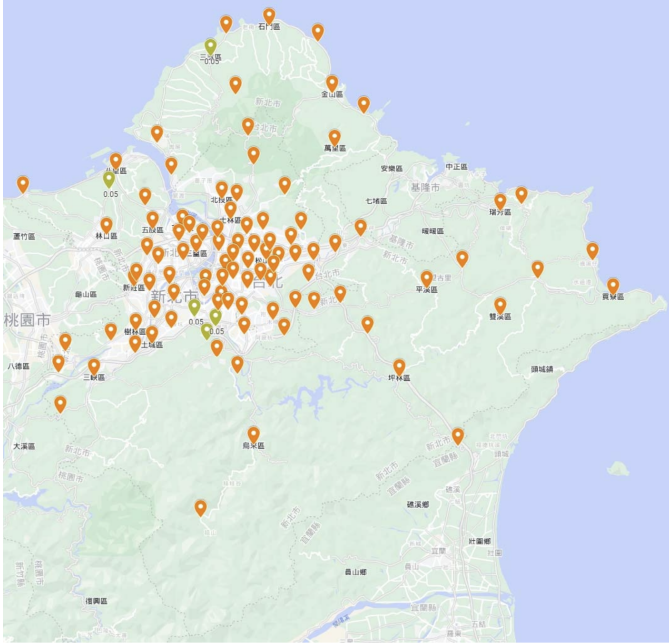
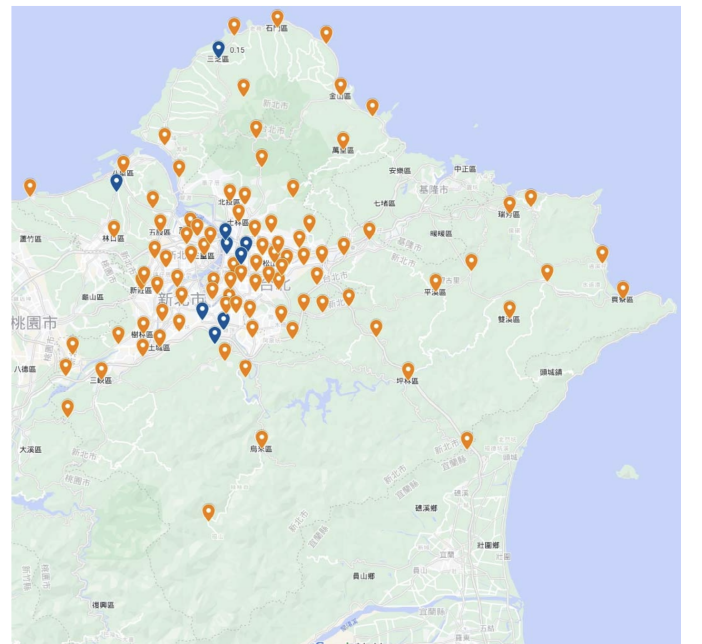


Fig. 5. Potential charging station locations [44].

Fig. 7. Charging station distribution map ($\alpha = 0.1$).Fig. 6. Charging station distribution map ($\alpha = 0.05$).Fig. 8. Charging station distribution map ($\alpha = 0.15$).

In summary, the proposed QUBO model combined with digital annealing outperforms classical optimization methods, demonstrating faster computation, lower cost solutions, and better scalability, particularly in large-scale and realistic urban settings.

VI. CONCLUSION

This paper introduces a model for positioning EV charging stations for solving the issue of limited driving range. The model is practical for real-world use, facilitating smooth EV travel. Three different algorithms to address this challenge are

evaluated and found that digital annealer is the most effective, especially for large-scale applications. The digital annealer is faster and slightly more accurate than the other methods, making it suitable for real-world scenarios and leading to a better setup of charging networks.

Regarding quantum computing, even though it's still in the early stages, our findings show that digital quantum computers are more effective than classical ones. As regular computing faces limitations and becomes less applicable, quantum computing is likely to take over in the near future. This is important for solving complex issues like setting up efficient EV charging stations and other real-world problems.

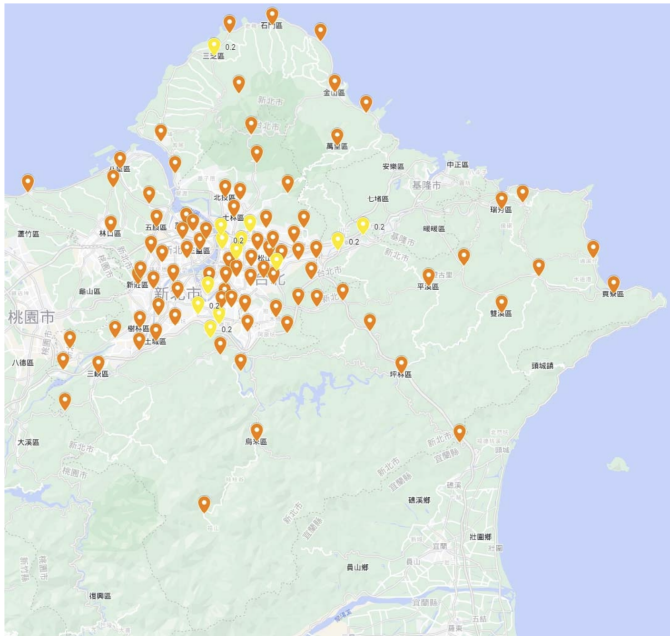


Fig. 9. Charging station distribution map ($\alpha = 0.2$).

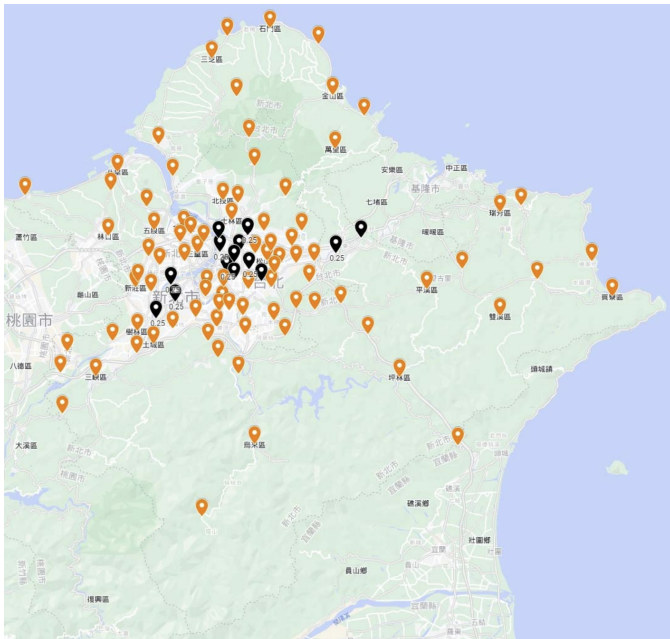


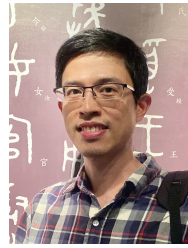
Fig. 10. Charging station distribution map ($\alpha = 0.25$).

As future work, we plan to extend our current approach to experiments on genuine quantum annealing hardware, such as the D-Wave quantum computer. Conducting these experiments will allow us to directly compare the performance and efficiency of digital annealing with true quantum annealing solutions. Such future research will further validate the quantum compatibility of our QUBO formulation, bridge the gap between quantum-inspired methods and quantum computing, and provide deeper insights into the practical advantages and limitations of actual quantum hardware in solving real-world optimization problems.

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